

# ML Specification Report — playground-series-s3e5

## Wine Quality — Ordinal Regression (Quadratic Weighted Kappa) • AIDE agent (20-step tree search, Claude Sonnet via shim)

All scores are taken from the run's `facts_aide.json`. Tree structure, per-step plans and per-node execution times come from `logs/2-camel-of-noble-might/journal.json`; hyperparameters from `best_solution.py` and the run's `config.yaml`; metric/target metadata from the competition `config.yaml`. Dataset shapes and the target's integer range are counted directly from the run's own `input/` directory.

**Superseded run.** This report documents AIDE run `2-camel-of-noble-might`, which the benchmark later **voided** for input contamination (see the input-purity note under Data) and replaced with a clean re-run, `0-eager-jovial-manul` (local OOF QWK 0.5747, 18 scored / 2 buggy, 1,152.1 s of execution; submitted 2026-07-29 for public 0.59362 / private 0.58138). Every figure below describes the voided run and is kept only as a record of it, including its zero-bug step count and its 209.5 s compute total. The benchmark's current s3e5 figures are the clean re-run's, and `benchmark_results/rerun_three_way.csv` records `winner_priv = aide` for this competition.

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## Overview

The task is to predict an integer wine `quality` score from eleven physicochemical measurements, scored by **quadratic weighted kappa (QWK, maximize)** — an ordinal metric that penalizes a prediction by the square of how many classes it misses by. AIDE searched it over **20 steps, all 20 of which executed and scored**, carrying a LightGBM-with-rounding draft at **0.49632** to a champion **local OOF QWK of 0.57654** at **step 19, the last step of the budget**. The champion is a single seed-bagged CatBoost regressor over an interaction-augmented feature set, whose predictions are converted to classes by a multi-start Nelder-Mead threshold optimizer.

The voided run scored **public 0.62064, private 0.59073, rank 73/903 — the 92.0nd percentile**, and the development-era score table recorded `local_winner = aide` with `lb_winner = mine` (0.59743), AIDE at 0.59073 ahead of the NVIDIA reproduce lane's 0.56974. That leaderboard verdict has since flipped the other way: on the current record the clean AIDE re-run's private 0.58138 beats the from-scratch lane's final-architecture re-run at 0.57223 and NVIDIA's 0.56974, and `benchmark_results/rerun_three_way.csv` records `winner_priv = aide`.

**Why it matters.** Ordinal targets with a squared-distance metric are their own problem class — sensory-quality grading, severity scoring, credit grades — and the decisive engineering choice is almost never the model but the map from a continuous score to discrete classes. This run is a clean demonstration: three quarters of AIDE's total gain came from replacing `round()` with learned cutpoints, and every subsequent model-side improvement was an order of magnitude smaller.

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## Data

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**Purpose of Data.** Predict an integer wine-quality grade from eleven continuous physicochemical features, with `quality` as target and `Id` as key, scored by QWK. **Data Format** is small, clean, all-numeric **tabular CSV**. **Data Volume**, counted from the run's `input/` directory, is **2,056 training rows × 13 columns** and **1,372 test rows × 12 columns** — eleven raw predictive features (`fixed_acidity`, `volatile_acidity`, `citric_acid`, `residual_sugar`, `chlorides`, `free_sulfur_dioxide`, `total_sulfur_dioxide`, `density`, `pH`, `sulphates`, `alcohol`).

**An input-purity note.** The `input/` directory also contained `train_processed.csv` ( $2,056 \times 23$ ) and `test_processed.csv` ( $1,372 \times 22$ ), which carry ten pre-engineered chemistry features on top of the raw eleven: `free_so2_ratio`, `bound_so2`, `fixed_to_volatile_acid`, `citric_to_volatile_acid`, `total_acid`, `alcohol_x_sulphates`, `alcohol_to_density`, `alcohol_x_density`, `sugar_to_alcohol`, `acid_ph_ratio`. AIDE read the processed files from step 0 onward and never touched `train.csv`, so its feature baseline is inherited rather than discovered; its own feature contribution is the automatic interaction block described below. The effect is milder than in s3e1 — the inherited block is ten simple ratios, not a full engineered pipeline — but it should be recorded.

**Data Quality** was never profiled (**no EDA artifact exists**). The one distributional fact AIDE acted on is class imbalance: it switched from `KFold` to `StratifiedKFold` at step 13 on the argument that "few very-low/very-high quality wines" make random folds unstable, and it tested inverse-frequency sample weighting at step 17 for the same reason. Measured from the run's input, the target takes six integer values, 3 through 8, and the two central classes dominate. Missing values, duplicates and train↔test drift were never measured by this run.

**Annotation Guidelines.** The label is an **integer grade in [3, 8]**, and the submission must contain integers — this is not a probability or ranking task. QWK's quadratic penalty means a 5-predicted-as-7 error costs four times a 5-predicted-as-6 error, which is precisely why the thresholding step dominates everything else.

**Feature Set.** The champion evaluates two feature sets and keeps the better:

```
23 (processed columns) - 2 (Id, quality) = 21 base features
C(6,2) = 15 top-feature pairs × 2 (product + ratio) = 30 generated features
21 + 30 = 51 features in the augmented arm
```

| Group                       | Origin                                                 | Features                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Raw physicochemical (11)    | competition data, via <code>train_processed.csv</code> | <code>fixed acidity</code> , <code>volatile acidity</code> , <code>citric acid</code> , <code>residual sugar</code> , <code>chlorides</code> , <code>free sulfur dioxide</code> , <code>total sulfur dioxide</code> , <code>density</code> , <code>pH</code> , <code>sulphates</code> , <code>alcohol</code>                   |
| Inherited engineering (10)  | present in <code>train_processed.csv</code>            | <code>free_so2_ratio</code> , <code>bound_so2</code> , <code>fixed_to_volatile_acid</code> , <code>citric_to_volatile_acid</code> , <code>total_acid</code> , <code>alcohol_x_sulphates</code> , <code>alcohol_to_density</code> , <code>alcohol_x_density</code> , <code>sugar_to_alcohol</code> , <code>acid_ph_ratio</code> |
| Automatic interactions (30) | AIDE, steps 18–19                                      | pairwise products and ratios among the six highest-importance base features, ranked by CatBoost gain from a first pass                                                                                                                                                                                                         |

The interaction block is generated, not designed: AIDE fits the baseline arm first, reads `get_feature_importance()`, takes the top six columns and emits every pairwise product and ratio. That is a mechanical mutation, and its measured worth is correspondingly small (see the trajectory).

**Splitting strategy.** `StratifiedKFold(n_splits=5, shuffle=True, random_state=42)`, stratified on the integer quality label, matching the harness's `k_fold_validation: 5`. Within each fold the model is trained three times with seeds 42, 123 and 7 and the predictions averaged, so the OOF vector is a seed-bagged continuous score. The plain `KFold` used up to step 12 was replaced at step 13 specifically to stabilize the rare extreme classes. The leaderboard was exercised: the champion's submission was scored publicly and privately.

## Models & Architecture

**Purpose of Architecture.** Produce a continuous latent quality score whose learned cutpoints maximize QWK. **Architecture Type** is a **single-learner regression-plus-thresholding pipeline**: one CatBoost regressor trained with squared-error loss on the integer target, seed-bagged three ways within each fold, followed by an `OptimizedRounder` that learns five cutpoints by direct QWK maximization. Notably, AIDE tried ensembling repeatedly and it never worked here — the LightGBM+XGBoost blend at step 5 scored 0.4082, well below the draft, and the top-3 hyperparameter-config average at step 16 lost to the single best config. The champion is deliberately a single model.

**Input Format** is a dense float matrix with no categorical columns. **Input Dimension** is **51 features** in the selected (augmented) arm, 21 in the baseline arm.

**Architecture Description.** CatBoost regressor with `depth 4`, `l2_leaf_reg 6.0`, `learning_rate 0.02`, `bagging_temperature 0.0`, `random_strength 1.0`, `iterations 2000`, `loss_function="RMSE"`, early stopping 100 on the fold's validation split with `use_best_model=True`. These values are the output of AIDE's own randomized search at steps 11–14, not defaults.

The `OptimizedRounder` is the second half of the architecture and deserves its own description. It holds a sorted vector of five cutpoints separating the six classes, applies them with `np.digitize`, and fits them by minimizing negative QWK under **Nelder-Mead** (`maxiter 10000`, `xatol / fatol 1e-6`). To escape local optima it runs **27 initializations** — a uniform half-integer start, a percentile-based start derived from the sorted prediction vector, and 25 Gaussian perturbations ( $\sigma = 0.3$ ) of the uniform start — and keeps whichever converged cutpoint set scores highest. Multi-start was itself an AIDE discovery: step 9 introduced it and moved the metric  $0.56841 \rightarrow 0.57275$ .

**Model Complexity** in fits:

```
2 feature arms × 3 seeds × 5 folds = 30 CatBoost fits
```

each with a 2000-iteration ceiling at depth 4, plus  $2 \times 27$  Nelder-Mead threshold optimizations. This is the smallest model in the batch by a wide margin, which is consistent with a 2,056-row training set.

## Training procedures

### Search trajectory

| Step | Parent | What AIDE changed                                      | OOF QWK |
|------|--------|--------------------------------------------------------|---------|
| 0    | —      | draft: LightGBM regression + simple rounding           | 0.49632 |
| 1    | —      | draft: LightGBM multiclass classification, argmax      | 0.48098 |
| 2    | 0      | <b>OptimizedRounder</b> — global Nelder-Mead cutpoints | 0.55672 |
| 3    | 2      | per-fold thresholds, averaged                          | 0.5452  |
| 4    | 2      | multi-seed bagging of LightGBM                         | 0.55917 |
| 5    | —      | draft: LightGBM + XGBoost ensemble                     | 0.4082  |
| 6    | —      | draft: CatBoost regression + global rounder            | 0.56841 |
| 7    | —      | draft: CatBoost + 3-seed bagging                       | 0.56591 |
| 8    | 6      | repeated $3 \times 5$ -fold CV                         | 0.56769 |
| 9    | 6      | multi-start threshold optimization (27 inits)          | 0.57275 |
| 10   | 9      | + seed bagging on top of multi-start                   | 0.56897 |
| 11   | 9      | randomized CatBoost hyperparameter search (9 configs)  | 0.5747  |
| 12   | 11     | MAE loss instead of RMSE                               | 0.56419 |
| 13   | 11     | StratifiedKfold instead of KFold                       | 0.57473 |

| Step | Parent | What AIDE changed                                                               | OOF QWK        |
|------|--------|---------------------------------------------------------------------------------|----------------|
| 14   | 13     | + <code>bagging_temperature</code> / <code>random_strength</code> in the search | 0.57589        |
| 15   | 14     | importance-based feature pruning (bottom 25 %) — rejected internally            | 0.57589        |
| 16   | 14     | top-3 config averaging — rejected internally                                    | 0.57589        |
| 17   | 14     | inverse-frequency sample weighting — rejected internally                        | 0.57589        |
| 18   | 14     | + generated pairwise interaction/ratio features                                 | 0.57596        |
| 19   | 18     | + 3-seed bagging inside each fold ( <b>champion</b> )                           | <b>0.57654</b> |

The tree opened with the configured **5 root drafts** (steps 0, 1, 5, 6, 7) and reached **depth 6**; the champion sits at depth 6 on the lineage 6 → 9 → 11 → 13 → 14 → 18 → 19, i.e. it descends from the fourth of the five drafts, not the first. **10 of the 20 scored steps set a new incumbent.** Total improvement:

$$0.57654 - 0.49632 = 0.08022 \quad (16.1630 \% \text{ relative})$$

**One change accounts for three quarters of the run.** Step 2 replaced round-to-nearest with learned cutpoints:

$$0.55672 - 0.49632 = 0.0604$$

$$0.0604 / 0.08022 = 75.29 \% \text{ of the run's total improvement}$$

Everything after step 2 — a better base learner, a hyperparameter search, stratification, generated interactions, seed bagging — bought the remaining quarter between them. AIDE's own report reaches the same conclusion ("thresholding matters more than model choice"), and it is the correct reading of this competition.

**The honest weakness is the plateau.** Steps 15, 16 and 17 all report **exactly 0.57589**, the step-14 incumbent's score. That is not a coincidence: each of those scripts tests its new lever internally (feature pruning, top-3 config averaging, class weighting), finds it worse, falls back to the incumbent configuration and reports the incumbent's number. So **4 of 20 steps (14 through 17) returned the same value**, 20 % of the budget spent confirming that three plausible levers do nothing. AIDE also spent two of its five root drafts on approaches that were worse than the first draft — multiclass classification at 0.48098 and the two-model ensemble at 0.4082, both below the 0.49632 baseline they were meant to improve on. On the other hand, this run wasted nothing on infrastructure: **zero buggy steps, zero timeouts**, and the whole search consumed 209.5 s.

A second, subtler weakness is in the metric itself. The reported 0.57654 is computed **after fitting the five cutpoints on the same OOF prediction vector that is then scored**, so it is not an unbiased estimate — the thresholds have seen the labels they are evaluated against. AIDE tested the cleaner alternative at step 3 (fit thresholds per fold, average them), measured 0.5452, and discarded it. With only five free parameters the optimism is small, and the leaderboard bears that out (private 0.59073 is above the local figure), but the model-selection objective was nonetheless partly self-referential.

One piece of context for reading "step 0": this journal is the second leg of the run, started with `initial_journal` pointing at a 5-node journal from an earlier leg, so AIDE's memory at step 0 already contained five prior attempts on this competition.

## Optimization specification

The **Loss Function** is squared error — CatBoost `loss_function="RMSE"` on the integer target treated as continuous — while **model selection is on QWK**, and the threshold layer optimizes QWK directly and non-differentiably. The two objectives are deliberately mismatched, and AIDE tested the alternative: step 12 switched the training loss to MAE and lost (0.56419 against the 0.5747 incumbent), confirming that squared error is the better latent-score objective even when the reported metric is kappa. The **Optimization Algorithm** is ordered gradient boosting for the base learner and **Nelder-Mead simplex search with 27 restarts** for the cutpoints. The **Learning Rate** is the boosting shrinkage, `learning_rate` 0.02 (the threshold optimizer has no learning rate — Nelder-Mead is derivative-free). There is **no Learning Rate Scheduler** (N/A — convergence is governed by early stopping at patience 100, not a schedule) and **no Batch Size** (N/A — full-dataset boosting).

**Training Duration** for the whole search was **209.5 s of executed wall-clock across 20 nodes** (mean 10.5 s, max 27.0 s), with no node coming within two orders of magnitude of the 1800 s harness timeout.

**Training Memory Consumption Limits** are **not recorded**.

**Transfer Learning** is N/A (no pretrained weights; the competition forbids them); its analogue is **journal memory**, and it is visibly the driver here — step 6's plan opens by reasoning that "the previous multi-model ensemble diluted the best single-model regression signal", which is what redirects the search from LightGBM to CatBoost and onto the winning lineage. **Data Augmentation** is N/A (tabular); the analogues are the generated interaction features and **seed bagging** (seeds 42, 123, 7 averaged within each fold).

**Reproducibility Standards** are moderate and unverified: `StratifiedKFold(random_state=42)`, model seeds fixed at [42, 123, 7], the threshold optimizer's perturbation RNG seeded at 42. But **AIDE runs no replay or rebuild gate** — 0.57654 is what one execution printed, and the champion was never re-run to confirm it. The multi-start Nelder-Mead fit is the component most likely to move under re-execution, since 25 of its 27 starts are random perturbations.

## Inference procedures

**Decision Threshold** is emphatically **not N/A here — it is the model's most important component**.

The CatBoost regressor emits a continuous score; the five fitted cutpoints partition it into the six classes by `np.digitize`, and the result is clipped to the observed label range [3, 8] and cast to `int`. The full inference chain for a test row is: average the fold-model predictions (test predictions are accumulated as `fold_test_preds / n_splits` across the five folds, each itself the mean of three seeds), apply the cutpoints learned on the OOF vector, clip, cast, then realign to `sample_submission.csv` id order. The submitted file therefore contains **integers, not scores**.

**Inference Duration** is **not recorded** separately — test scoring happens inside the fold loop. **Inference Memory Consumption Limits** are **not recorded**; the whole pipeline is a 2,000-row problem.

## Evaluation & Benchmarking

**Task for Performance Evaluation.** Assign an integer quality grade to each of the 1,372 test wines and submit a two-column CSV in `sample_submission.csv` order. **Performance Metrics.** QWK, maximized. AIDE's local trajectory runs  $0.49632 \rightarrow 0.55672 \rightarrow 0.56841 \rightarrow 0.57275 \rightarrow 0.5747 \rightarrow 0.57589 \rightarrow 0.57654$  across the draft, the threshold optimizer, the CatBoost switch, multi-start thresholds, the hyperparameter search, stratification and the final augmented seed-bagged champion. Local and leaderboard agree in direction and the local estimate is conservative:

```
0.59073 (private) - 0.57654 (local 00F) = 0.01419
0.62064 (public) - 0.59073 (private) = 0.02991
```

The 0.02991 public-to-private spread is the widest in the batch relative to the metric's scale, which is expected for a kappa computed on 1,372 test rows — QWK is a high-variance statistic when the extreme grades are rare, which is the premise AIDE itself acted on when it switched to stratified folds at step 13. The submission placed **73/903, the 92.0nd percentile**.

**Performance Benchmarking.** All three lanes submitted to the same private leaderboard:

| Lane     | Approach                                                     | Private QWK    | Note                                                                       |
|----------|--------------------------------------------------------------|----------------|----------------------------------------------------------------------------|
| My agent | from-scratch GBDT pool + tree search                         | <b>0.59743</b> | <code>lb_winner</code> in the development-era table; superseded, see below |
| AIDE     | 20-step tree search, CatBoost + multi-start OptimizedRounder | 0.59073        | behind by 0.0067                                                           |
| NVIDIA   | reproduce-agent, public-kernel replication                   | 0.56974        | behind AIDE by 0.02099                                                     |

```
0.59743 - 0.59073 = 0.0067
0.59073 - 0.56974 = 0.02099
```

AIDE loses the episode by 0.0067 — a margin roughly the size of a single one of its own late-stage steps — while beating the NVIDIA reproduce lane by more than three times that. `facts_aide.json` records `local_winner = aide` and `lb_winner = mine`; the other lanes' local numbers are outside this report's source scope, so the disagreement is reported, not resolved. Both private figures in the table belong to the voided run and the pre-re-run from-scratch lane. On the current benchmark record the clean AIDE re-run scores private 0.58138 against that lane's 0.57223, and `benchmark_results/rerun_three_way.csv` awards s3e5 to AIDE (`winner_priv = aide`).

The instructive part is where AIDE's remaining gap sits. It found the decisive idea (learned cutpoints) at step 2 out of 20 and then spent eighteen steps extracting 0.02 more from a single CatBoost, four of them on a flat plateau and two of its five drafts on approaches strictly worse than the baseline. It never

revisited the threshold layer after step 9 — no per-class cost analysis, no ordinal-native objective, no calibration of the latent score — even though its own evidence said that layer was where the value was. A tree search with a good local move and a weak sense of which lever has remaining headroom will convert its budget into diminishing single-model tuning, which is exactly what the last six steps show.